

Program : Diploma in Biomedical Engineering	
Course Code : 6242D	Course Title: IoT in Healthcare
Semester : 6	Credits: 4
Course Category : Open Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester : 60

Course Objectives:

- To provide an introduction to the internet of things to an understanding level.
- To cover the concepts of IoT protocols, prototyping boards including Arduino and Raspberry pi, sensors actuators, and protocol modules.
- To introduce domain-specific applications of IoT in hospital automation, electronic patient record, patient monitoring systems.

Course Prerequisites:

Topic	Course code	Course name	Semester
IT skills		Introduction to IT systems	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concept of Internet of Things.	14	Understanding
CO2	Illustrate Arduino-based prototyping boards and supporting protocols.	14	Understanding
CO3	Demonstrate interfacing of sensors and actuators with Arduino.	14	Understanding
CO4	Illustrate healthcare applications of IoT.	14	Understanding

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concept of Internet of Things.		
M1.01	Differentiate Internet of Things and machine to machine communication.	4	Understanding
M1.02	Explain architecture and IoT platforms	5	Understanding
M1.03	List various challenges in IoT	5	Understanding
Contents: Definition - Internet of Things, Machine to Machine (M2M) communication. M2M vs. IOT; Introduction to IOT- Understanding IoT fundamentals, IOT Architecture, Various Platforms for IoT; Challenges in IOT			
CO2	Illustrate Arduino-based prototyping boards and supporting protocols.		
M2.01	Interpret microcontrollers.	2	Understanding
M2.02	Explain Arduino based prototyping boards	4	Understanding
M2.03	Demonstrate Raspberry pi and its configuring	4	Understanding
M2.04	Classify various IoT protocols and modules.	4	Understanding
	Series Test – I	2	
Contents: Prototyping boards: Microcontrollers - block diagram, example for microcontrollers; Prototyping boards - Arduino Uno, Nano, NodeMCU - block diagram, concept of shield; Raspberry Pi - Specifications, Block diagram, OS installation procedures; Protocols - Serial, Bluetooth, Wifi, Zigbee - specification, application; Basics of Wireless Networking - HC06 Bluetooth module, ESP8266 Wi-Fi Module			

CO3	Demonstrate interfacing of sensors and actuators with Arduino.		
M3.01	Classify analog and digital sensors	4	Understanding
M3.02	Interface sensors with Arduino.	5	Understanding
M3.03	Interface relays and motors with Arduino	5	Understanding

Contents:

Sensor & Actuators - overview of Sensors working, Analog and Digital Sensors, Interfacing of Temperature, Humidity, Motion, Light and Gas Sensor with Arduino, Interfacing of Actuators with Arduino - Relay, Switch, dc motor and Servo Motor.

CO4	Illustrate domain-specific applications of IoT.		
M 4.01	Demonstrate hospital automation.	5	Understanding
M 4.02	Explain the application of IoT in healthcare.	4	Understanding
M 4.03	Explain the application of IoT in patient monitoring.	4	Understanding
	Series Test – II	2	

Contents:

Domain specific IoT: Introduction, Hospital automation- inventory control in hospitals, IoT based electronic medical records.

IoT in healthcare monitoring - basic block diagram, IoT based ECG monitoring, Epilepsy monitoring, IoT in rehabilitation

Text / Reference

T/R	Book Title/Author
T1	The Internet of Things – Key Applications and Protocols, Wiley Publication, Olivier Hersent, David Boswarthick, Omar Elloumi. ISBN: 9788126557653
T2	The Internet of Things , Pearson, By Michael Miller ISBN: 9789332552456
R1	Smart Medical Data Sensing and IoT Systems Design in Healthcare (2019), Chinmay Chakraborty, IGI Global

Online Resources:

Sl.No	Website Link
1	Arduino resources available at https://www.arduino.cc/
2	Arduino compatibles available at https://www.adafruit.com/
3	Getting started with Raspberry Pi, available at https://projects.raspberrypi.org/